

Ripple

An NLP risk engine that turns news and social posts into calibrated risk signals, and stress-tests a wholesale banking book the moment a high-impact event breaks.

0.75

event macro-F1 of the distilled model, vs 0.57 for keyword rules

47%

of impact ≥ 7 alerts followed by a $\geq 2\sigma$ abnormal price move (base rate 11%)

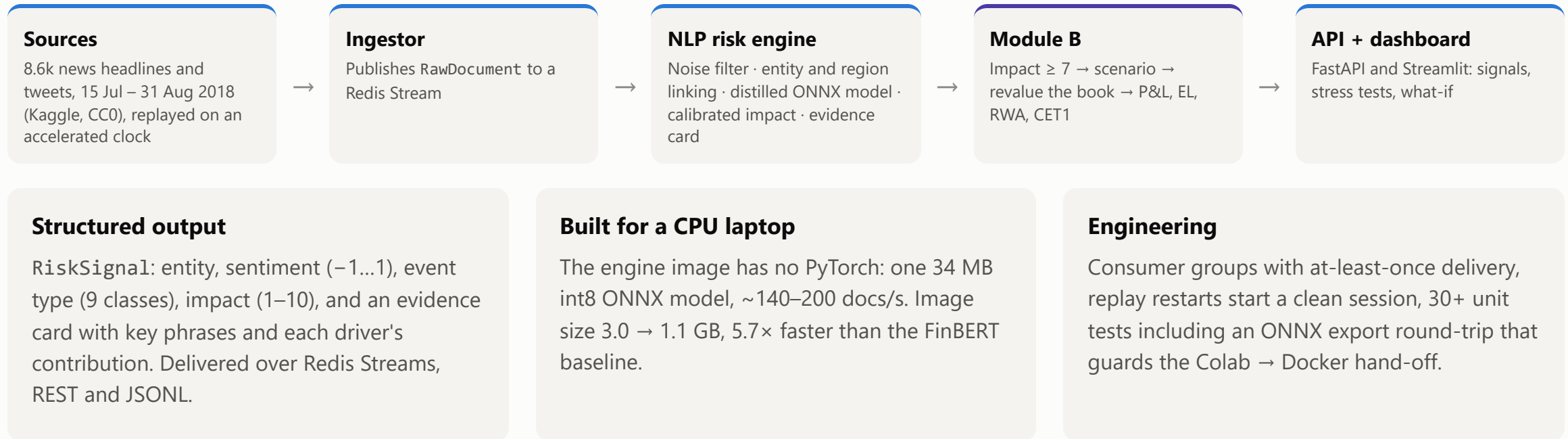
98

automatic stress tests in the 7-week replay; worst: $-\$49\text{m}$ (lira crisis), CET1 to 11.7%

1 cmd

`docker compose up`: six services, offline, no API keys

Six services on an event stream, reproducible end to end

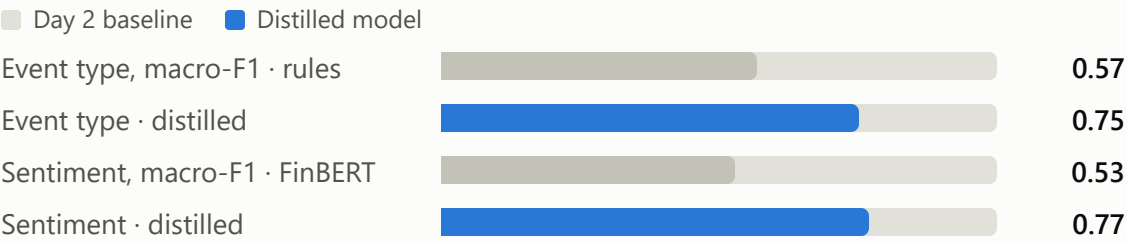


A 7B LLM's judgement, distilled into a 33M model that runs on CPU

Teacher: Qwen2.5-7B-Instruct (open weights) labelled 20.8k headlines and tweets with event type, sentiment and severity on a Colab L4 GPU.

Student: bge-small (33M parameters) fine-tuned with three heads, exported to int8 ONNX.

Honest evaluation: the demo window (4,941 docs) was held out of training. Scores are agreement with the teacher, not human labels.



Data traps found and fixed

91% of company-tagged tweets are spam or chatter	Noise filter turns a buried signal into a clear spike on Facebook's crash day
The news set's ticker column is not the subject	"Facebook's Dismal Q2" was tagged ADI and BUD; entities are linked from the text
The tweet set's returns are wrong	+23% for Facebook on its –19% day; returns recomputed from prices
Mixed date formats silently dropped 96% of one file's dates	Found during calibration; the demo now has twice the news
Market news named a region but scored against the S&P 500	Region entities (Turkey, China, ...) with their own buzz and market

Impact means something: calibrated to what markets actually did

Impact estimates **P(material move)**: a 2-day market-adjusted return of at least 2σ for that stock or region. A logistic model, fitted on 129k historical signals (2014–2020, demo window excluded), using severity, sentiment, event type, buzz and source.

Impact ≥ 7 , the stress-test trigger, means the top 3% of historical signals by predicted move probability.

Every signal shows its drivers in log-odds:

Facebook (FB) Q2 Earnings Miss, Shares Fall on Dim Outlook · impact 9.9 · P(move) 60%

Drivers (log-odds): buzz +2.24 · earnings event +0.78 · severity +0.77 · sentiment +0.02

Findings: **buzz** (abnormal attention) is the strongest driver; sentiment adds almost nothing to *how big* a move is but sets its *direction*, which Module B uses.

Test on the held-out demo window

	HAND-SET FORMULA	CALIBRATED
Alert precision (impact $\geq 7 \rightarrow \geq 2\sigma$ move)	32%	47%
Stock-day ranking AUC	0.645	0.615
Validation 2019–20 AUC (28k signals)	0.661	0.655
Spearman with abnormal move	0.144	0.154
Base rate of $\geq 2\sigma$ moves		11%

On ranking, calibration only matches a well-tuned hand-set formula: predicting moves from headlines alone is hard. What it adds is meaning: impact is a probability, the alert threshold is the top 3% of history, and alerts are $\sim 4\times$ more likely than chance to precede a 2σ move.

High-impact events trigger a revaluation of a wholesale book

Book: 97 positions, \$12.6bn: loans to 51 borrowers (the 30 tracked companies plus corporates in Turkey, China, Europe, India and LatAm), corporate and sovereign bonds, rate swaps, FX forwards, CDS, equity swaps. Synthetic and seeded: the brief's retail transaction data does not describe a wholesale book.

Scenarios measured from real episodes (Yahoo Finance), scaled by impact (×40% at 7, full episode at 10):

SIGNAL	REPLAYS	KEY SHOCKS
Geopolitical	Trade war, May 2019	SPY −4.5%, CNY −1.3%, HY +48bp
Sovereign / EM credit	Lira crisis, Aug 2018	TRY −28%, EM +43bp
Macroeconomic	Taper tantrum, 2013	10y +92bp, EM +115bp
Company event	Issuer-specific	−k·σ of that stock, spread +10–400bp

Pricing: duration and convexity, DV01, spread DV01, delta, FX translation; loan EL with spread-stressed PDs; **Basel IRB** RWA and CET1.

Full-severity results (impact 10)

EVENT	P&L	CET1
Rates shock (taper tantrum)	−\$279m	13.0% → 6.8%
Turkish lira crisis	−\$116m	13.0% → 10.2%
US–China trade war	−\$81m	13.0% → 10.8%
Facebook earnings miss	−\$7.5m	13.0% → 12.8%

The book's biggest vulnerability is **rate duration** (\$1.6bn of sovereign bonds), not credit. Hedges behave: the short-TRY forward offsets the FX loss on lira loans; the pay-fixed swap offsets part of the bond losses.

It stays calm on chatter and fires on what moved prices: Facebook's crash and the lira crisis

8,556

documents ingested; 953 spam and chatter posts filtered as noise

7,395

risk signals; 234 high-impact (≥ 7)

98

stress tests triggered automatically

11.7%

lowest stressed CET1 ratio (from 13.0%)

TIME (2018)	TRIGGER HEADLINE	ENTITY · EVENT	IMPACT	SCENARIO	P&L	CET1
20 Jul 07:45	Microsoft (MSFT) Beats Q4 Earnings, Top Line Grows on Azure	MSFT · Earnings	8.0	Issuer shock (positive news)	+\$0.8m	13.02%
25 Jul 16:38	Plunging Facebook takes a chunk out of the QQQs	FB · Earnings	8.5	Issuer shock	-\$5.2m	12.86%
26 Jul 17:41	Facebook (FB) Q2 Earnings Miss, Shares Fall on Dim Outlook	FB · Earnings	9.9	Issuer shock, re-run on escalation	-\$7.5m	12.81%
10 Aug 16:09	Trump's Turkey tariffs seen raising costs for Kinder Morgan...	TURKEY · Geopolitical	7.0	Turkish lira crisis ×40%	-\$49.3m	11.74%
16 Aug 04:00	Walmart Soars On Q2 Earnings Beat, Other Retailers Follow	WMT · Earnings	9.2	Issuer shock (positive; no WMT equity)	\$0.0m	13.00%

What we would tell a risk committee

Limitations, stated plainly

- NLP scores measure agreement with an LLM teacher, not human analysts; a human-labelled test set is the next step.
- Impact predicts move size modestly (AUC in the mid-0.6s); it is a triage signal, not a trading signal.
- The portfolio is synthetic; regional CDS widening is assumed where free data does not exist.
- Pricing is first-order (sensitivities), CET1 covers credit RWA only, losses are pre-tax.
- Replay of 2018 data, not a live feed.

Next steps

- Live ingestion (GDELT, RSS) behind the same RawDocument contract.
- Reverse stress test: find the smallest scenario that breaches a CET1 floor.
- Rating-migration matrices instead of PD scaling; full revaluation for options.
- Module A: feed signals into an ESG/risk-tilted index rebalancer.
- Human-in-the-loop: analysts confirm or reject alerts, and the labels retrain the student.

Everything here is open source and reproducible: **docker compose up** runs the full demo offline; scripts rebuild every dataset, label, model and calibration from public data.